

From Teacher Pathways to Invariant Manifolds: Consensus Subspace Distillation for TSFMs

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Abstract

Time-series foundation models (TSFMs) deliver strong cross-domain generalization, but their scale makes deployment costly. Knowledge distillation is a natural compression route, yet prior TSFM distillation typically imitates teacher outputs, features, or pairwise relations, and therefore remains tightly coupled to teacher-specific training trajectories while underutilizing two empirical properties: (i) high-level representations across model scales tend to converge toward a shared, approximately low-rank geometry, and (ii) layer-wise utility follows a long-tail pattern. We propose *consensus subspace distillation*, which reframes distillation as aligning a student to a model-agnostic geometric object: a scale-invariant low-rank consensus subspace together with its center statistics. Offline, we screen high-contribution layers via drop-layer marginal loss, estimate a shrinkage-stabilized covariance from their embeddings, and derive a truncated eigensubspace that defines a consensus projector. Online, we project student embeddings into this subspace and match the teacher’s projected mean and covariance using a lightweight mean–covariance objective, enabling stable optimization without rigid pointwise feature binding. To mitigate subset-induced bias, we further introduce a frequency-domain uncertainty injection mechanism that inflates spectral density based on characteristic-function discrepancies and injects dispersion only within the consensus directions. Across forecasting and imputation, the distilled student matches or slightly improves upon the teacher, while exhibiting a predictable trade-off under strict zero-shot classification. With MOMENT-Large as teacher, we

achieve about 90% parameter reduction and substantial distillation-time savings while retaining comparable performance across multiple time-series tasks. Code and compressed weights are available at github.com/icml-csd/CSD.

1. Introduction

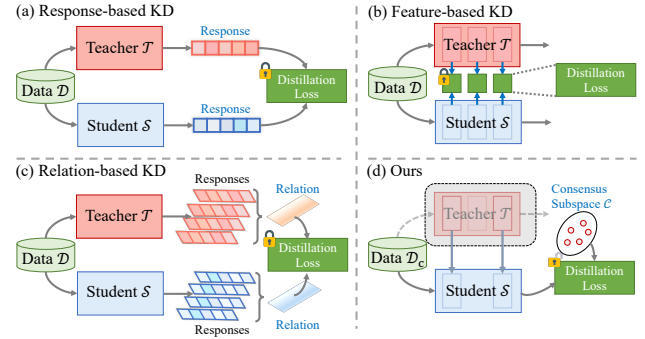


Figure 1. Comparison of four knowledge distillation (KD) paradigms. (a) Response-based distillation aligns only output probabilities and ignores dynamic information in intermediate layers. (b) Feature-based distillation enforces activation matching but disrupts natural convergence processes. (c) Relation-based distillation focuses on inter-sample similarity yet overlooks cross-layer self-organizing consistency. (d) Our proposed consensus subspace optimization method extracts scale-invariant low-rank subspaces to guide student models toward geometric centers. This avoids dependence on teacher-specific pathways.

Transformer-based time series foundation models (TSFMs) have significantly advanced the processing of complex sequential data. These models enable multitask generalization and robust predictions across various domains (Liang et al., 2024). However, as model scales grow, the associated computational and storage overheads rise substantially. This limits their deployment in resource-constrained environments. To address this challenge, model compression techniques have become essential. They compress large TSFMs into efficient versions while preserving performance as much as possible (Liu et al., 2025; Shi et al., 2025).

Among compression strategies for TSFMs, several approaches stand out for reducing model size and inference

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costs. These include neural architecture search (NAS), pruning, knowledge distillation (KD), quantization, and low-rank mapping (Fournier et al., 2023). NAS automatically designs efficient architectures, though it often involves high search costs (Wang et al., 2026). Pruning simplifies models by removing redundant weights, but it may impair the representational capacity of critical hierarchical structures (Xu et al., 2022). Quantization reduces numerical precision for compression, yet improper tuning can compromise generalization on long sequences (Li et al., 2024). Low-rank mapping captures key information subspaces, but it often overlooks hierarchical imbalances in Transformers (Zha et al., 2024). In contrast, knowledge distillation transfers knowledge from teacher models to student models effectively. It maintains consensus representations and addresses hierarchical imbalances in TSFMs. This makes it a promising choice for efficient knowledge transfer without training from scratch.

A core challenge in distilling TSFMs lies in efficiently transferring hierarchical knowledge while preserving the model’s inherent characteristics. Traditional distillation methods fall into three categories: response-based, feature-based, and relation-based. Response-based KD aligns output logits or soft labels to emphasize probabilistic imitation (Hinton et al., 2015). However, it often overlooks dynamic information in intermediate layers (see Fig. 1(a)). Feature-based KD matches intermediate activations, but this rigid binding disrupts natural convergence to scale-invariant subspaces and amplifies low-level noise (Fig. 1(b)) (Romero et al., 2014; Zhu & Zhang, 2025). Relation-based KD focuses on inter-sample or inter-layer similarities, such as attention maps (Park et al., 2019a). Yet it neglects the self-organizing alignment of high-level embeddings across model scales. This leads to low distillation efficiency and insufficient generalization, especially when layer contributions follow long-tail distributions (Fig. 1(c)).

These limitations become evident in empirical studies. During masked autoencoder (MAE) pretraining on time series, high-level embeddings from models of varying scales, such as 12-layer and 24-layer, tend to converge to consistent consensus subspaces. They show minimal unconstrained MAE differences (see Fig. 2(a)) and highly aligned projection centers (Fig. 2(b)). However, existing methods fail to leverage this phenomenon. Instead, they bind student models to teacher-specific paths and overlook the independent contributions of shallow layers to context capture (Fig. 2(c)). This results in amplified biases and slower convergence.

Unlike response/feature KD that enforces pointwise imitation, or relation KD that matches pairwise structures, we distill a *model-agnostic* geometric object: a scale-invariant low-rank consensus subspace together with its center statistics. This shifts the supervision target from teacher-specific

pathways to an invariant manifold shared across scales, enabling aggressive compression with stable optimization.

To overcome these issues, we propose a novel KD framework that redefines distillation as a consensus space optimization task. This approach exploits the spontaneous convergence of high-level embeddings to scale-invariant, low-rank subspaces across model scales (see Fig. 1(d)). Specifically, we apply singular value decomposition on consensus layers over local data subsets to extract low-rank subspaces and construct projection matrices. These guide student representations to align with the teacher’s geometric centers, enabling efficient knowledge transfer without disrupting natural convergence structures. Unlike the rigid matching in feature distillation or the limitations in relation distillation, our method prioritizes subspace consensus over pointwise matching. It decouples from the teacher model, accelerating convergence and enhancing robustness across scales.

Our method also incorporates a scalable uncertainty injection mechanism to bridge biases from data selection. This models biases as frequency-domain gaps and enhances generalization to unseen data through inflated covariances. The design draws from two key insights. First, consensus spaces form spontaneously at varying depths, with deeper layers adding redundancy without altering geometric centers (Fig. 2(a-b)). Second, layer contributions are uneven, with shallow layers exhibiting zero-shot reconstruction capabilities and deeper layers showing long-tail redundancy (Fig. 2(c)). Our main contributions include:

- A new distillation perspective that transforms KD into a consensus space optimization problem, leveraging scale-invariant subspaces and hierarchical long-tail distributions for efficient TSFM compression.
- A method combining decomposition-based projection with frequency-domain uncertainty injection to align student representations, mitigate biases, and improve generalization.
- Across forecasting and imputation, the distilled student matches or slightly improves upon the teacher with over 90% parameter reduction, while exhibiting a predictable trade-off under strict zero-shot classification (Fig. 3).

2. Related Work

2.1. Efficient Time Series Foundation Modeling

Recent studies have emphasized Transformer-based time series foundation models (TSFMs) for handling complex data and enabling multi-task generalization (Liang et al., 2024). Masked reconstruction (MR), drawing inspiration

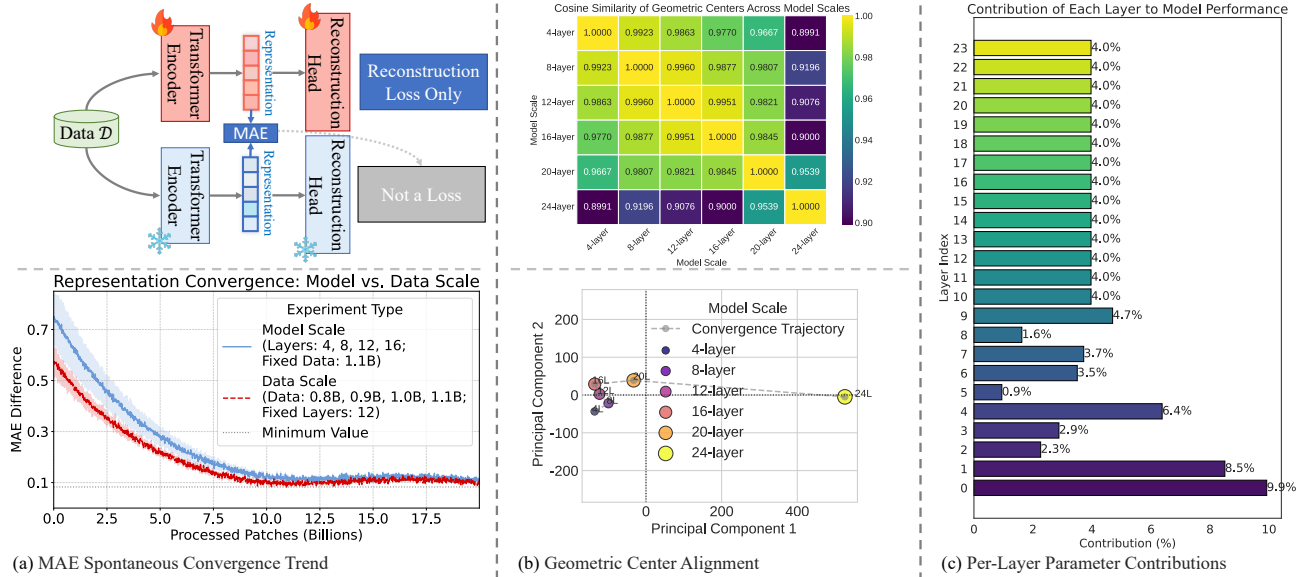


Figure 2. Empirical motivation validation. (a) During pre-training of time series foundation models across varying data and model scales, representations spontaneously converge to align with those of larger pre-trained models under unconstrained conditions. The upper panel illustrates the convergence measurement methodology, while the lower panel shows results with error bands across scales. (b) Projected representations reveal geometrically aligned centers across scales. The upper panel displays a cosine similarity heatmap of geometric centers, indicating highly aligned spaces. The lower panel depicts centroid offset trajectories in a reference frame (using the 24-layer model’s center as origin), demonstrating tight clustering and convergence via PCA projection. Together, (a) and (b) suggest a potential consensus subspace with an invariant geometric center. (c) Bar chart of per-layer parameter contributions, showing a long-tail distribution.

from large language models, allows these models to learn robust representations. This approach works by randomly masking and reconstructing sequences, which helps capture contextual dependencies and long-term patterns (Liu et al., 2025). Notable examples include MOMENT, which uses multi-dataset pre-training to enhance data diversity (Goswami et al., 2024). PatchTST employs patching and channel-independent processing to enable efficient long-sequence forecasting (Nie et al., 2023). Time-LLM reprograms large language models with MR and prompt engineering, thereby improving generalization (Jin et al., 2024). Additionally, Time-MoE scales up to 2.4 billion parameters, validating scaling laws across various domains (Shi et al., 2025). Despite these advances, the increasing model sizes create significant computational and storage challenges. This motivates our research on TSFM compression.

Various compression techniques exist, such as neural architecture search (NAS), pruning, knowledge distillation (KD), quantization, and low-rank mapping (Fournier et al., 2023). For instance, NAS methods like those proposed by Wang et al. optimize multivariate architectures but often require high computational resources and offer limited adaptability (Wang et al., 2026). Xu et al. introduced contrastive pruning to preserve task-agnostic knowledge through contrastive learning, which enhances generalization in pre-trained models (Xu et al., 2022). However, this approach overlooks high-level consensus in TSFMs. Quantization techniques,

such as GPTQ (Frantar et al., 2023), perform well on proxy tasks but struggle with uncertainty calibration, which can impair long-context generalization (Li et al., 2024). Low-rank mapping reduces dimensions for key representations. Zha et al. (2024) applied decoupled spatio-temporal compression to high-dimensional data, yet it ignores hierarchical long-tail distributions, limiting its applicability to TSFMs. In comparison, KD effectively transfers knowledge from teacher to student models. This method better preserves TSFM consensus spaces and addresses hierarchical imbalances (Gao et al., 2024; Wu et al., 2021).

2.2. Knowledge Distillation in TSFMs

Knowledge distillation can be categorized into response-based methods, which imitate the teacher’s soft label outputs (Hinton et al., 2015), feature-based methods, which align intermediate representations to capture deeper knowledge (Romero et al., 2014), and relation-based methods, which mine inter-instance or inter-layer relationships, such as similarity matrices, for robust transfer (Park et al., 2019a). For compressing Transformer-based TSFMs, several approaches have emerged. These include self-distillation frameworks that improve self-supervised efficiency through masked view prediction (Pieper et al., 2023), task-specific distillation with adversarial augmentation for downstream robustness (Zhang et al., 2022), and methods inspired by neural collapse or low-rank local feature distillation for better representation align-

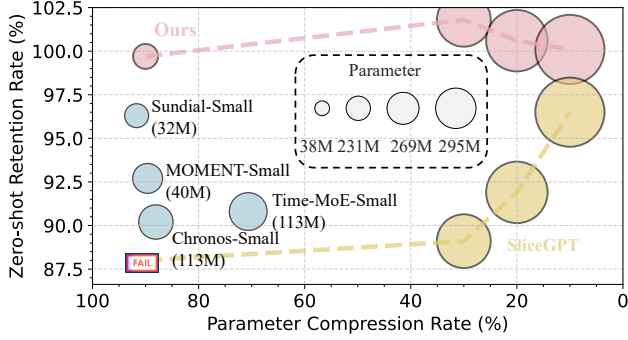


Figure 3. Benchmarked against MOMENT-Large, our method surpasses state-of-the-art approaches in zero-shot long-horizon forecasting retention and model compression. Unlike conventional methods that fail to significantly compress pre-trained models while preserving high-quality representations, our approach achieves a 90.13% parameter reduction. Even compared to the smallest mainstream time series foundation models, our compressed model delivers extreme compression while approximately preserving the original large-scale model performance.

ment (Zhang et al., 2025; Sy et al., 2025). Other techniques involve SliceGPT, which removes rows or columns from weight matrices to achieve up to 25% parameter reduction while preserving zero-shot performance (Ashkboos et al., 2024), FrameQuant’s dynamic bit adjustment for balancing accuracy and efficiency (Adepu et al., 2024), probabilistic knowledge transfer for deep representation learning (Pasalis & Tefas, 2018), relation mining (Park et al., 2019b), and contrastive distillation for enhanced alignment (Tian et al., 2020).

Existing methods often overlook the intrinsic low-rank consensus and uneven layer contributions in TSFM embeddings, leading to suboptimal compression (Ni et al., 2026; Zhao et al., 2024; Xu et al., 2023). We address this by reformulating distillation as consensus space optimization, leveraging inherent model structures to guide efficient compression.

3. Methodology

In this section, we introduce a novel consensus subspace distillation framework, as depicted in Fig. 1(d). This framework integrates scale-invariant low-rank representations, leveraging inherent model-agnostic structures observed across diverse time series foundation models (TSFMs), with hierarchical contribution screening to enable efficient compression. We then describe an extensible uncertainty injection mechanism to mitigate subset bias and enhance generalization. Finally, we outline the training procedure and associated losses.

Notation. Let an input time series be $x \in \mathbb{R}^{C \times L}$ with C channels and length L . After tokenization (e.g., patching), we obtain T tokens. The Transformer hidden size is H .

For a batch of size B , the l -th layer token embeddings are $E^l \in \mathbb{R}^{B \times T \times H}$ (we omit model subscripts when clear). When forming covariances, we reshape to $\bar{E}^l \in \mathbb{R}^{H \times (BT)}$.

3.1. Consensus Subspace Distillation

Offline Computation For a given teacher model $\mathcal{T}$ and the full dataset $\mathcal{D} \in \mathbb{R}^{N \times C \times L}$, we first follow the central limit theorem to randomly sample an $n\%$ subset $\mathcal{D}_c \in \mathbb{R}^{\tilde{N} \times C \times L}$ for offline statistical estimation.

High-contribution layer screening. To identify layers that contribute most to reconstruction, we estimate a *drop-layer marginal loss* on the subset $\mathcal{D}_c$. Specifically, let $\mathcal{T}_{\setminus l}$ denote the teacher with the l -th Transformer block bypassed via an identity skip (keeping dimensions consistent). We define

$$\Delta \mathcal{L}^l = \mathcal{L}_{\text{MAE}}(\mathcal{T}_{\setminus l}) - \mathcal{L}_{\text{MAE}}(\mathcal{T}),$$

and select the top- K indices with the largest $\Delta \mathcal{L}^l$ as $\mathcal{I}_{\text{top}}$. This procedure is stable in practice and reflects the marginal utility of each block under the original parameterization.

Next, we construct the consensus subspace: for each selected layer $l \in \mathcal{I}_{\text{top}}$, reshape the embedding $\bar{E}_T^l = \text{reshape}(E_T^l) \in \mathbb{R}^{H \times (\tilde{N}T)}$, where E_T^l is the token embedding from the l -th layer of $\mathcal{T}$ (with T tokens), and perform mean subtraction $\mu_l = \frac{1}{\tilde{N}T} \bar{E}_T^l \mathbf{1}_{\tilde{N}T}$, $\tilde{E}_T^l = \bar{E}_T^l - \mu_l \mathbf{1}_{\tilde{N}T}^\top$. Compute the average covariance

$$\Sigma_0 = \frac{1}{K} \sum_{l \in \mathcal{I}_{\text{top}}} \frac{1}{\tilde{N}T} \tilde{E}_T^l \tilde{E}_T^{l\top} \in \mathbb{R}^{H \times H}. \quad (1)$$

Since for any non-zero $v \in \mathbb{R}^H$, $v^\top \bar{\Sigma} v = v^\top \Sigma_0 v + \gamma \|v\|_2^2 \geq \gamma \|v\|_2^2 > 0$, to ensure positive definiteness, we apply shrinkage to obtain $\bar{\Sigma} = \Sigma_0 + \gamma I_H$ ($\gamma > 0$).

Then, perform $\bar{\Sigma} = U \Lambda U^\top$, where $\Lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_H)$. For rank truncation, select the rank based on the cumulative variance threshold θ :

$$r = \min \left\{ m : \frac{\sum_{i=1}^m \lambda_i}{\text{tr } \Lambda} \geq \theta \right\}, \quad U_r = [u_1, \dots, u_r]. \quad (2)$$

The consensus projection matrix is $P_c = U_r U_r^\top \in \mathbb{R}^{H \times H}$. For the token embedding E_T^M from the top layer of $\mathcal{T}$, project $\bar{E}_T = P_c E_T^M$, where M is the total number of layers in $\mathcal{T}$. Reshape $Z = \text{reshape}(\bar{E}_T) \in \mathbb{R}^{H \times (\tilde{N}T)}$ and compute

$$\begin{cases} \mu_T = \frac{1}{\tilde{N}T} Z \mathbf{1}_{\tilde{N}T}, \\ \Sigma_T = \frac{1}{\tilde{N}T} (Z - \mu_T \mathbf{1}_{\tilde{N}T}^\top) (Z - \mu_T \mathbf{1}_{\tilde{N}T}^\top)^\top. \end{cases} \quad (3)$$

Computation of Spectral Density $S(\omega_k)$ In the offline computation phase, we obtain $S(\omega_k)$ for subsequent uncertainty injection. Based on the projected teacher embedding $\tilde{E}_T \in \mathbb{R}^{\tilde{N} \times C \times T}$, first reshape it to $Z \in \mathbb{R}^{H \times (\tilde{N}T)}$. Perform FFT along the token dimension T to obtain the frequency-domain embedding $E_{T,f} \in \mathbb{C}^{\tilde{N} \times C \times T}$. For each frequency point $\omega_k = \frac{2\pi k}{T}$ ($k = 0, 1, \dots, T-1$), compute the frequency-domain covariance matrix of the embeddings:

$$S(\omega_k) = \frac{1}{\tilde{N}C} \sum_{n=1}^{\tilde{N}} \sum_{c=1}^C \left[\text{Re}(E_{T,f,n,c,k}) \cdot \text{Re}(E_{T,f,n,c,k})^\top + \text{Im}(E_{T,f,n,c,k}) \cdot \text{Im}(E_{T,f,n,c,k})^\top \right] \in \mathbb{R}^{H \times H}. \quad (4)$$

The teacher cache provides the baseline spectral density $S(\omega_k) \in \mathbb{R}^{H \times H}$, which serves as the frequency-domain statistical benchmark of the teacher model on the subset $\mathcal{D}_c$, effectively capturing the covariance between spatial dimensions after embedding.

Student Network Initialization Copy the K layers corresponding to $\mathcal{I}_{\text{top}}$ from the teacher $\mathcal{T}$ to the student $\mathcal{S}$, and add a low-rank increment only to the MLP output weights W_{mlp} of the copied layers:

$$W_{\text{mlp}}^{\text{new}} = W_{\text{mlp}} + \mathbf{1}_{[r_a > 0]} \underbrace{AB}_{\text{rank } r_a}, \quad (5)$$

where $A \in \mathbb{R}^{H \times r_a}$, $B \in \mathbb{R}^{r_a \times H}$, and $r_a \ll H$. During training, freeze the original weights and only update A and B (if $r_a = 0$, it degenerates to an identity mapping).

Remark: The construction of the consensus subspace $\mathcal{C}$ ensures scale invariance, as dimensionality reduction captures the low-rank structure of embeddings, which spontaneously converges across different models, supporting efficient subspace alignment.

Next, we introduce the mean-covariance alignment loss for training consensus distillation.

Mean-Covariance Alignment Loss For the student top-layer embedding E_S^K , project $\tilde{E}_S = P_C E_S^K$, reshape $Z = \text{reshape}(\tilde{E}_S) \in \mathbb{R}^{H \times (BT)}$, and compute

$$\begin{cases} \mu_S = \frac{1}{BT} Z \mathbf{1}_{BT}, \\ \Sigma_S = \frac{1}{BT} (Z - \mu_S \mathbf{1}_{BT}^\top)(Z - \mu_S \mathbf{1}_{BT}^\top)^\top. \end{cases} \quad (6)$$

For two Gaussian distributions $\mathcal{N}(\mu_T, \Sigma_T)$ and $\mathcal{N}(\mu_S, \Sigma_S)$, their squared 2-Wasserstein distance satisfies:

$$\mathcal{W}_2^2(\mathcal{N}_T, \mathcal{N}_S) = \underbrace{\|\mu_T - \mu_S\|_2^2}_{\mathcal{L}_\mu} + \underbrace{\text{tr}\left(\Sigma_T + \Sigma_S - 2(\Sigma_T^{1/2} \Sigma_S \Sigma_T^{1/2})^{1/2}\right)}_{\Phi(\Sigma_S)}. \quad (7)$$

The covariance term $\Phi(\Sigma_S)$ has complex gradient computation, so we introduce a surrogate objective $g(\Sigma_S) = \|\Sigma_S - \Sigma_T\|_F^2$, as g shares the global minimum with Φ at $\Sigma_S = \Sigma_T$. Thus, the objectives are $\mathcal{L}_\mu = \|\mu_S - \mu_T\|_2^2$ and $\mathcal{L}_\Sigma = g(\Sigma_S)$.

Our method unifies distillation, subspace projection, and mean-covariance alignment within a single framework, introducing a latent anchor manifold—rather than relying on a single teacher—based on new insights into scale-invariant convergence in TSFMs.

3.2. Uncertainty Injection

We quantify the gap using frequency-domain ChF differences, converting it into spectral density inflation noise injected solely into the consensus subspace. This leverages uneven layer contributions to enhance conservatism without disrupting core representation centers, extending static alignment to dynamic uncertainty augmentation for improved generalization on unseen data. Because inflation is applied only within $\text{span}(U_r)$, the injection increases dispersion along consensus directions while preserving the orthogonal complement, preventing the student from being pushed toward teacher-specific off-manifold trajectories.

Uncertainty computation directly uses the original input sequences $\mathcal{D} \in \mathbb{R}^{N \times C \times L}$ without requiring teacher forward passes, avoiding additional computational overhead. Perform fast Fourier transform (FFT) on the L axis (time dimension) of the sequences to obtain the frequency-domain representation $x_f \in \mathbb{C}^{N \times C \times L}$ (complex tensor). Similarly, batch the subset $\mathcal{D}_c \in \mathbb{R}^{\tilde{N} \times C \times L}$ to obtain $\tilde{x}_f \in \mathbb{C}^{B \times C \times L}$. The frequency points ω_k are defined as the discrete frequencies in standard FFT: $\omega_k = \frac{2\pi k}{L}$, where $k = 0, 1, \dots, L-1$.

ChF Difference Quantification Inspired by Liang et al. (2024), we estimate the empirical characteristic function (ChF) to quantify the statistical differences between the two distributions $\mathcal{D}$ and $\mathcal{D}_c$ in the frequency domain. The ChF is the Fourier transform of the distribution, similar to a probability generating function, used to capture higher-order moment deviations. Computations are performed on the raw sequences, averaging across the spatial dimension C to integrate multivariate information:

$$\Phi_x(\omega_k) = \frac{1}{NC} \sum_{n=1}^N \sum_{c=1}^C e^{i \text{Re}(x_{f,n,c,k})}, \quad (8)$$

where $\Phi_x(\omega_k)$ represents the empirical ChF of samples x in $\mathcal{D}$ at frequency ω_k . Similarly, for samples $\tilde{x}$ in $\mathcal{D}_c$, $\Phi_{\tilde{x}}(\omega_k) = \frac{1}{BC} \sum_{b,c} e^{i \text{Re}(\tilde{x}_{f,b,c,k})}$.

Then, compute the ChF difference to quantify the distance

Table 1. Comparison of different model compression methods. Results are averaged over multiple datasets for each task. Long-Horizon Forecasting uses average MAE over 8 datasets with forecast horizons {96, 192, 336, 720}; Imputation uses average MAE over 6 datasets with mask ratios {0.125, 0.25, 0.375, 0.5}; Anomaly Detection uses average Adj. Best F1 and VUS-ROC over 248 datasets; Classification uses average accuracy over 91 datasets.

Method	Long-Horizon Forecasting $\downarrow$	Imputation $\downarrow$	Anomaly Detection $\downarrow$		Classification $\uparrow$	Comp. Time $\downarrow$	Model Parameters	Ref.
	MAE (Avg.) $\downarrow$	MAE (Avg.) $\downarrow$	Adj. Best F1 $\uparrow$	VUS-ROC $\uparrow$				
Teacher	0.476	0.159	0.721	0.728	0.764	N/A	385M	ICML'24
Ours	0.471 (+1.05%)	0.157 (+1.26%)	0.713 (-1.11%)	0.721 (-0.96%)	0.633 (-17.01%)	~ 28 h	38M (-90.13%)	-
SliceGPT	0.528 (-10.92%)	0.176 (-10.69%)	0.630 (-12.62%)	0.647 (-11.13%)	0.685 (-10.34%)	~ 10 s	274M (-28.83%)	ICLR'24
FrameQuant	0.585 (-22.90%)	0.218 (-37.11%)	0.541 (-25.00%)	0.572 (-21.43%)	0.580 (-24.08%)	~ 3 h	385M (15x Mem. $\downarrow$)	ICML'24

Table 2. Comparison of different distillation methods under 9.87% parameter retention (student model). We can achieve promising results using only 10% of the data for distribution distillation. Results are averaged over multiple datasets for each task. Long-Horizon Forecasting uses MAE averaged over 8 datasets with forecast horizons 96, 720. Imputation uses MAE averaged over 6 datasets with mask ratios 0.125, 0.500. Anomaly Detection uses Adj. Best F1 and VUS-ROC averaged over 248 datasets. Classification uses accuracy averaged over 91 datasets. Distillation Time indicates the time taken for the distillation process.

Method	Long-Horizon Forecasting $\downarrow$		Imputation $\downarrow$		Anomaly Detection $\downarrow$		Classification $\uparrow$	Distillation Time (GPU h) $\downarrow$	Ref.
	MAE (96) $\downarrow$	MAE (720) $\downarrow$	MAE (0.125) $\downarrow$	MAE (0.500) $\downarrow$	Adj. Best F1 $\uparrow$	VUS-ROC $\uparrow$			
Teacher	0.299	0.381	0.159	0.158	0.569	0.660	0.764	N/A	ICML'24
PKT	0.418 (-39.80%)	0.514 (-34.91%)	0.203 (-27.67%)	0.229 (-44.94%)	0.478 (-16.00%)	0.561 (-15.00%)	0.550 (-28.01%)	204.73	ECCV'18
RKD	0.368 (-23.08%)	0.457 (-19.95%)	0.183 (-15.09%)	0.198 (-25.32%)	0.512 (-10.02%)	0.594 (-10.00%)	0.588 (-23.04%)	180.19	CVPR'19
CRD	0.344 (-15.05%)	0.434 (-13.91%)	0.171 (-7.55%)	0.187 (-18.35%)	0.535 (-5.98%)	0.620 (-6.06%)	0.610 (-20.16%)	220.46	ICLR'20
ADA-KD	0.359 (-20.07%)	0.465 (-22.05%)	0.190 (-19.50%)	0.205 (-29.75%)	0.524 (-7.91%)	0.607 (-8.03%)	0.595 (-22.12%)	190.82	AAAI'22
LRLFD	0.353 (-18.06%)	0.449 (-17.85%)	0.175 (-10.06%)	0.193 (-22.15%)	0.518 (-8.96%)	0.600 (-9.09%)	0.602 (-21.20%)	210.37	NAACL'25
NCKD	0.347 (-16.05%)	0.442 (-16.01%)	0.179 (-12.58%)	0.200 (-26.58%)	0.529 (-7.03%)	0.613 (-7.12%)	0.605 (-20.81%)	195.64	AAAI'25
Ours	0.287 (+4.01%)	0.366 (+3.94%)	0.152 (+4.40%)	0.151 (+4.43%)	0.557 (-2.11%)	0.647 (-1.97%)	0.633 (-17.15%)	3.86	-

between the two ChFs:

$$\begin{aligned} \text{Chf}(\omega_k) &= |\Phi_x(\omega_k)|^2 + |\Phi_{\bar{x}}(\omega_k)|^2 \\ &\quad - 2|\Phi_x(\omega_k)||\Phi_{\bar{x}}(\omega_k)|\cos(a_x - a_{\bar{x}}), \end{aligned} \quad (9)$$

where $|\Phi_x(\omega_k)|$ is the magnitude of $\Phi_x(\omega_k)$, and $a_x = \arg \Phi_x(\omega_k)$ is the argument. The global uncertainty $\mathcal{U}$ aggregates differences across all frequencies:

$$\mathcal{U} = \sum_{k=0}^{L-1} \text{Chf}(\omega_k)w(\omega_k), \quad w(\omega_k) = \exp(-\omega_k^2/2\sigma^2), \quad (10)$$

where $w(\omega_k)$ is a Gaussian weighting function.

Spectral Density Inflation First, based on the ChF differences, construct a scaling factor (inflation factor):

$$\gamma(\omega_k) = 1 + \lambda \frac{\text{Chf}(\omega_k)}{|\Phi_x(\omega_k)|^2 + \varepsilon}, \quad 0 < \lambda \leq 1. \quad (11)$$

Then, compute the inflated spectral density $S_\star(\omega_k) = \gamma(\omega_k)S(\omega_k)$. Further, obtain the zero-lag covariance:

$$\Sigma_\star = \frac{1}{T} \sum_{k=0}^{T-1} S_\star(\omega_k). \quad (12)$$

If $\mathcal{U} \rightarrow 0$, then $\Sigma_\star \rightarrow \Sigma_T$.

Consensus Space Injection Project $\Sigma_\star^C = U_r^\top \Sigma_\star U_r \in \mathbb{R}^{r \times r}$. Perform eigenvalue decomposition on $\Sigma_\star^C$ to get

$\Sigma_\star^C = V_\star \Lambda_\star V_\star^\top$, where $\Lambda_\star = \text{diag}(\lambda_{\star,1} \geq \dots \geq \lambda_{\star,r})$ is the diagonal matrix of eigenvalues. Then, compute the gain coefficients:

$$\Gamma = \text{diag}\left(\sqrt{\frac{\lambda_{\star,1}}{\lambda_1}}, \dots, \sqrt{\frac{\lambda_{\star,r}}{\lambda_r}}\right), \quad (13)$$

where λ_i are from the original consensus subspace's Λ . The final uncertainty-injected consensus projection matrix is $P_C^\gamma = U_r \Gamma U_r^\top$. These formulations, though complex, precisely capture scale-invariant subspaces, supported by empirical convergence in Fig. 2.

3.3. Overall Procedure

In the offline phase, precompute P_C , (μ_T, Σ_T) , $\{S(\omega_k)\}$, $\Sigma_\star$, and Γ . In the online phase, the student computes E_S^K , obtains $\bar{E}_S = P_C^\gamma E_S^K$ via the uncertainty-injected consensus projection, and feeds it into the mean-covariance alignment loss. The overall loss is $\mathcal{L} = \mathcal{L}_{\text{task}} + \beta(\mathcal{L}_\mu + \mathcal{L}_\Sigma)$, with gradients backpropagated naturally. The algorithmic procedure is summarized in Algorithm 1.

Complexity. Offline computation forms an $H \times H$ covariance and performs an eigen-decomposition once, with $O(H^2 \tilde{N}T + H^3)$ time and $O(H^2)$ memory. Online distillation adds only a projection and mean/covariance statistics per batch, i.e., $O(H^2 BT)$ in the naive form, and can be implemented efficiently using streaming covariance updates;

Algorithm 1 Consensus Subspace Distillation Framework

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1: Input:  $\mathcal{T}, \mathcal{D}, n\%, K, \theta, \lambda, \sigma, \beta$ .
2: Output:  $\mathcal{S}$ .
3: Offline:
4: Sample  $\mathcal{D}_c$  from  $\mathcal{D}$ .
5: Select  $\mathcal{I}_{\text{top}}$  by  $\Delta\mathcal{L}^l$ .
6: Build  $P_c$  from  $\bar{\Sigma}$ .
7: Compute  $(\mu_T, \Sigma_T), S(\omega_k), \mathcal{U}$ .
8: Inflate to  $P_c^\gamma$ .
9: Init: Copy top layers to  $\mathcal{S}$ .
10: Online:
11: while not converged do
12:   Get  $E_S^K$ , project to  $\bar{E}_S$ .
13:   Compute losses:  $\mathcal{L} = \mathcal{L}_{\text{task}} + \beta(\mathcal{L}_\mu + \mathcal{L}_\Sigma)$ .
14:   Update  $\mathcal{S}$ .
15: end while
    
```

no additional teacher forward pass is required during uncertainty computation.

4. Experiment Results

4.1. Dataset and Model Setup

Dataset Selection We evaluate the proposed consensus subspace distillation framework on the Time Series Pile, a diverse collection of approximately 13 million time series spanning 13 distinct real-world domains, including healthcare, electricity, economics, and transportation, with a total of 1.23 billion timestamps (Goswami et al., 2024). This dataset ensures comprehensive validation of the model’s cross-domain generalization capabilities. Following the experimental setup of MOMENT, we select datasets for long-term forecasting, imputation, classification, and anomaly detection tasks. We adopt standard dataset splits and pre-processing procedures, with detailed metadata and splitting methods available in our code repository.

Evaluation protocol. We report (i) **zero-shot** performance by directly applying the pre-trained backbone without any task-specific gradient updates, and (ii) **linear probing (LP)** where only a lightweight linear head is trained while the backbone is frozen. We keep the same pre-processing, horizons/mask ratios, and splits as MOMENT, and apply identical protocols to all baselines for fair comparison.

Baselines and Teacher Model. We use MOMENT-Large as the teacher model (Goswami et al., 2024), pretrained on the Time Series Pile through masked reconstruction. The baselines include state-of-the-art knowledge distillation methods: Probabilistic Knowledge Transfer (PKT) (Passalis & Tefas, 2018), Relational Knowledge Distillation (RKD) (Park et al., 2019b), Contrastive Representation Distillation

(CRD) (Tian et al., 2020), Adversarial Data Augmentation for KD (ADA-KD) (Zhang et al., 2022), Low-Rank Local Feature Distillation (LRLFD) (Sy et al., 2025), and Neural Collapse Inspired KD (NCKD) (Zhang et al., 2025). We also include SliceGPT (Ashkboos et al., 2024), which supports low-rank mapping and pruning, along with the quantization method FrameQuant (Adepu et al., 2024).

4.2. Implementation Details

All experiments use 8 NVIDIA RTX 4090 GPUs, with distillation on one 4090 GPU for 3 epochs. We copy the top $K = 3$ layers from the teacher model, selected via marginal contribution screening, and add a low-rank increment to MLP output weights with rank $r_a = 64$. In the offline phase, a $n = 10\%$ dataset subset is sampled to compute the consensus subspace projection matrix P_c , using a truncation threshold of $\theta = 0.99$ and spectral density inflation ($\lambda = 0.1, \sigma = 1.0$). In the online phase, we use the AdamW optimizer (learning rate 1×10^{-4} , batch size $B = 2048$) and loss weighting $\beta = 0.5$. Full implementation details, including random seeds, training configurations, ablation study protocols, and model weights, are available in our public code repository.

4.3. Compare with SOTA Methods

Compression Performance Comparison We benchmark our consensus subspace optimization method against MOMENT-Large (385M parameters) and other state-of-the-art techniques, such as SliceGPT and FrameQuant (Tab. 1). Our approach delivers comparable performance in forecasting and imputation, while retaining over 90% accuracy in anomaly detection and classification tasks. Our method compresses the model to 38M parameters, achieving a 90.13% reduction that significantly surpasses the 28.83% from SliceGPT. By extracting scale-invariant low-rank subspaces, our technique overcomes rigid matching challenges and validates that consensus spaces serve as effective manifolds for efficient knowledge transfer.

Distillation Method Comparison Tab. 2 shows our method outperforms prior distillation techniques using only 9.87% of the parameters and 3.86 GPU hours of training. It improves upon the teacher model with a 4.01% MAE reduction in forecasting and a 4.40% reduction in imputation, while maintaining robust performance in anomaly detection and classification. These results demonstrate that our low-rank alignment to geometric centers enables efficient and bias-resistant knowledge transfer.

Foundation Model Comparison The analysis in Tab. 3 demonstrates that our compressed model sustains competitive performance against larger time series foundation models. It excels in datasets like ETTm1 and ETTh2 with lower

Table 3. Zero-shot long-horizon forecasting comparison between our compressed model and mainstream time series foundation models. Corresponding prediction lengths include {96, 192, 336, 720}. Averaged results of four prediction lengths are reported here. 1st Count refers to the number of datasets where the current model attains the top-ranked average performance over all forecasting horizons. Results of baseline models are officially reported by Liu et al. (2025). Datasets in pre-training are not evaluated on corresponding models, which are denoted by the dash (—).

Models	Ours		MOMENT _{Small}		MOMENT _{Large}		Time-MoE _{Base}		Time-MoE _{Large}		Time-MoE _{Ultra}		Sundial _{Small}		Sundial _{Base}		Sundial _{Large}		Chronos _{Base}		Chronos _{Large}	
	-		ICML'24		ICML'24		ICLR'25		ICLR'25		ICLR'25		ICML'25		ICML'25		ICML'25		TMLR'24		TMLR'24	
Metric↓	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETtm1	0.356	0.381	0.354	0.391	0.345	0.380	0.394	0.415	0.376	0.405	0.356	0.391	0.354	0.388	0.336	0.377	0.331	0.369	0.645	0.500	0.555	0.465
ETtm2	0.262	0.318	0.269	0.325	0.260	0.319	0.317	0.365	0.316	0.361	0.288	0.344	0.265	0.324	0.258	0.320	0.254	0.315	0.310	0.350	0.295	0.338
ETTh1	0.403	0.436	0.427	0.442	0.419	0.435	0.400	0.424	0.394	0.419	0.412	0.426	0.390	0.418	0.411	0.434	0.395	0.420	0.591	0.468	0.588	0.466
ETTh2	0.351	0.397	0.358	0.411	0.353	0.395	0.366	0.404	0.405	0.415	0.371	0.399	0.340	0.387	0.333	0.387	0.334	0.387	0.405	0.410	0.455	0.427
ECL	0.169	0.270	0.171	0.264	0.166	0.261	-	-	-	-	-	-	0.169	0.265	0.169	0.265	0.166	0.262	0.214	0.278	0.204	0.273
Weather	0.226	0.269	0.237	0.277	0.227	0.268	0.265	0.297	0.270	0.300	0.256	0.288	0.233	0.271	0.234	0.270	0.238	0.275	0.292	0.315	0.279	0.306
1 st Count	37	36	26	23	40	44	16	16	16	16	19	22	41	40	43	40	48	48	3	5	7	8

Method	Anomaly Detection $\uparrow$		Classification $\uparrow$
	Adj. Best F1 $\uparrow$	VUS-ROC $\uparrow$	
Teacher	0.723	0.726	0.767
Ours ($K=4$)	0.651 (-9.89%)	0.662 (-9.38%)	0.704 (-8.36%)
Ours ($K=4$, w/o UI)	0.645 (-10.79%)	0.656 (-10.19%)	0.698 (-9.00%)
Ours ($K=8$)	0.717 (-1.54%)	0.721 (-1.16%)	0.752 (-1.87%)
Ours ($K=8$, w/o UI)	0.712 (-1.94%)	0.716 (-1.93%)	0.747 (-2.48%)
Ours ($K=12$)	0.728 (-0.12%)	0.729 (-0.43%)	0.761 (-0.54%)
Ours ($K=12$, w/o UI)	0.724 (-0.69%)	0.725 (-0.96%)	0.757 (-1.17%)
Ours ($K=16$)	0.716 (-0.89%)	0.727 (-0.81%)	0.758 (-1.12%)
Ours ($K=16$, w/o UI)	0.711 (-1.66%)	0.722 (-1.52%)	0.753 (-1.83%)

Table 4. Ablation study on the number of layers (K) in our method, including cases without uncertainty injection (UI).

average MSE and MAE. These results emphasize balanced retention of temporal dynamics through subspace alignment, even after significant parameter reduction.

4.4. Ablation Study

Tab. 4 presents our ablation on the number of layers (K). Higher values, such as $K = 12$, produce student models that nearly match the teacher’s performance in anomaly detection and classification. This indicates that deeper hierarchical integration better captures essential consensus subspaces. Variants with uncertainty injection (UI) consistently outperform those without, particularly at lower K , by compensating for representational biases and enabling robust knowledge transfer. The trend suggests an optimal K around 12, where further increases add redundancy without proportional benefits in alignment to scale-invariant geometric centers.

4.5. Does Consensus-Space Distillation Preserve Discriminative Representations?

Fig. 4 visualizes the learned representations on the large-scale Crop classification dataset, comparing our method

Second-Best Distillation Method Our Distillation Method

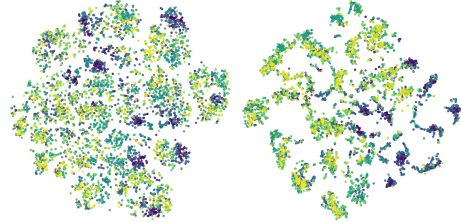


Figure 4. PCA and t-SNE visualizations of the representations learned by our method and other baselines on the Crop datasets, with distinct colors indicating different classes. Without dataset-specific fine-tuning, our method produces separable, clustered representations, indicating potential for effective feature extraction in downstream classification.

with CRD, the second-best approach. Our method yields more distinct and well-separated class representations, even in a zero-shot setting without ground-truth labels. The representation is from the distilled model’s final layer output.

5. Limitations and Discussion

Our approach assumes the existence of a stable low-rank consensus subspace across model scales; this may weaken for architectures with fundamentally different tokenization or normalization schemes, or under extreme domain shifts where representation geometry changes substantially. In addition, spectral inflation introduces hyperparameters (λ , σ) whose optimal values can vary with sampling ratio $n\%$ and sequence length. We provide sensitivity analyses and failure cases in the appendix.

6. Conclusion

We reformulate knowledge distillation for Transformer-based time series foundation models as a consensus subspace optimization problem, exploiting the convergence

of high-level embeddings to scale-invariant, low-rank subspaces. Our framework achieves 90.13% parameter reduction and 100x distillation speedup while maintaining performance across zero-shot tasks including forecasting, imputation, anomaly detection, and classification. Experiments demonstrate superiority over state-of-the-art methods, enabling efficient deployment in resource-limited settings.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Adepu, H., Zeng, Z., Zhang, L., and Singh, V. FrameQuant: Flexible Low-Bit Quantization for Transformers. In *Proceedings of the International Conf. on Machine Learning (ICML)*, pp. 203–227. PMLR, 2024.
- Ashkboos, S., Croci, M. L., do Nascimento, M. G., Hoefler, T., and Hensman, J. SliceGPT: Compress Large Language Models by Deleting Rows and Columns. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2024.
- Fournier, Q., Caron, G. M., and Aloise, D. A practical survey on faster and lighter transformers. *ACM Computing Surveys*, 55(14s):1–40, 2023.
- Frantar, E., Ashkboos, S., Hoefler, T., and Alistarh, D. Optq: Accurate quantization for generative pre-trained transformers. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2023.
- Gao, H., Ren, Q., and Li, J. Distillation enhanced time series forecasting network with momentum contrastive learning. *Information Sciences*, 675:120712, 2024.
- Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., and Dubrawski, A. MOMENT: a family of open time-series foundation models. In *Proceedings of the International Conf. on Machine Learning (ICML)*, pp. 16115–16152. PMLR, 2024.
- Hinton, G., Vinyals, O., and Dean, J. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*, 2015.
- Jin, M., Wang, S., Ma, L., Chu, Z., Zhang, J., Shi, X., Chen, P.-Y., Liang, Y., Li, Y.-f., and Pan, S. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2024.
- Li, S., Ning, X., Wang, L., Liu, T., Shi, X., Yan, S., Dai, G., Yang, H., and Wang, Y. Evaluating quantized large language models. In *Proceedings of the International Conf. on Machine Learning (ICML)*, pp. 28480–28524. PMLR, 2024.
- Liang, Y., Wen, H., Nie, Y., Jiang, Y., Jin, M., Song, D., Pan, S., and Wen, Q. Foundation models for time series analysis: A tutorial and survey. In *Proceedings of the ACM SIGKDD Conf. on Knowledge Discovery and Data Mining (KDD)*, pp. 6555–6565, 2024.
- Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J., and Long, M. Sundial: A Family of Highly Capable Time Series Foundation Models. In *Proceedings of the International Conf. on Machine Learning (ICML)*, 2025.
- Ni, J., Liu, Z., Wang, S., Jin, M., and Jin, W. Timedis: Efficient long-term time series forecasting with mlp via cross-architecture distillation. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*, pp. 1113–1124. Association for Computing Machinery, 2026.
- Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2023.
- Park, W., Kim, D., Lu, Y., and Cho, M. Relational knowledge distillation. In *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, pp. 3967–3976, 2019a.
- Park, W., Kim, D., Lu, Y., and Cho, M. Relational knowledge distillation. In *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, pp. 3967–3976, 2019b.
- Passalis, N. and Tefas, A. Learning deep representations with probabilistic knowledge transfer. In *European Conf. on Computer Vision (ECCV)*, pp. 268–284, 2018.
- Pieper, F., Ditschuneit, K., Genzel, M., Lindt, A., and Otterbach, J. Self-distilled representation learning for time series. *arXiv preprint arXiv:2311.11335*, 2023.

- Romero, A., Ballas, N., Kahou, S. E., Chassang, A., Gatta, C., and Bengio, Y. Fitnets: Hints for thin deep nets. *arXiv preprint arXiv:1412.6550*, 2014.
- Shi, X., Wang, S., Nie, Y., Li, D., Zhou, Y., Wen, Q., and Jin, M. Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2025.
- Sy, Y., Cerisara, C., and Illina, I. Efficient One-shot Compression via Low-Rank Local Feature Distillation. In *Proceedings of the Conf. of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pp. 5643–5661, 2025.
- Tian, Y., Krishnan, D., and Isola, P. Contrastive Representation Distillation. In *Proceedings of the International Conf. on Learning Representations (ICLR)*, 2020.
- Wang, Y., Wu, H., Dong, J., Liu, Y., Wang, C., Long, M., and Wang, J. Deep time series models: A comprehensive survey and benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–20, 2026.
- Wu, Z., Jiang, Y., Zhao, M., Cui, C., Yang, Z., Xue, X., and Qi, H. Spirit distillation: A model compression method with multi-domain knowledge transfer. In *International Conference on Knowledge Science, Engineering and Management*, pp. 553–565. Springer, 2021.
- Xu, Q., Wu, M., Li, X., Mao, K., and Chen, Z. Distilling Universal and Joint Knowledge for Cross-Domain Model Compression on Time Series Data. In *Proceedings of the International Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 4460–4468, 2023.
- Xu, R., Luo, F., Wang, C., Chang, B., Huang, J., Huang, S., and Huang, F. From dense to sparse: Contrastive pruning for better pre-trained language model compression. In *Proceedings of the AAAI Conf. on Artificial Intelligence (AAAI)*, volume 36, pp. 11547–11555, 2022.
- Zha, R., Zhang, L., Li, S., Zhou, J., Xu, T., Xiong, H., and Chen, E. Scaling up multivariate time series pre-training with decoupled spatial-temporal representations. In *Proceedings of the IEEE International Conf. on Data Engineering (ICDE)*, pp. 667–678. IEEE, 2024.
- Zhang, M., Naresh, N. U., and He, Y. Adversarial data augmentation for task-specific knowledge distillation of pre-trained transformers. In *Proceedings of the AAAI Conf. on Artificial Intelligence (AAAI)*, volume 36, pp. 11685–11693, 2022.
- Zhang, S., Song, Z., and He, K. Neural collapse inspired knowledge distillation. In *Proceedings of the AAAI Conf. on Artificial Intelligence (AAAI)*, 2025.
- Zhao, S., Jin, M., Hou, Z., Yang, C., Li, Z., Wen, Q., and Wang, Y. Himtm: Hierarchical multi-scale masked time series modeling with self-distillation for long-term forecasting. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pp. 3352–3362, 2024.
- Zhu, Z. and Zhang, W. Exploring feature-based knowledge distillation for recommender system: A frequency perspective. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, pp. 2182–2193, 2025.